

MIMO Beamforming Experimental Results and Reproducibility Report

Extracted from the completed seed-42 verification run

1. Scope and provenance

This report packages the numerical outputs extracted from the completed verification experiment generated with the GitHub-ready reproducibility runner. The executed run used seed 42 and the medium-quick configuration. It is a reproducibility record, not final multi-seed statistical evidence.

2. Executed configuration

Parameter	Value
N	4 antennas
NB	16 beam/codebook entries
Training SNR	10 dB
Episodes	100
Steps/episode	20
Batch size	32
Gamma	0.95
Learning rate	0.001
Epsilon decay	0.995
Mobility	0.05
Bandwidth	10 MHz
Target update	every 10 episodes
Evaluation	30 fresh channels
Evaluation SNR grid	-10, -5, 0, 5, 10, 15, 20 dB
Seed	42

3. Final training summary

Algorithm	Rate mean	Rate std	BER mean	Capacity (Mbps)
DQN	1.861082	0.805530	0.047475	18.459915
Double DQN	1.715141	0.830384	0.049968	17.829244
Dueling DQN	2.019297	0.648347	0.026248	19.807305

4. SNR sweep

DQN

SNR	Agent	Oracle	Gap	BER
-10	0.048330	0.082367	0.034037	0.402702
-5	0.183648	0.254247	0.070599	0.307705
0	0.502749	0.708215	0.205466	0.191512

5	0.965987	1.520290	0.554303	0.102818
10	2.061673	2.805952	0.744279	0.038199
15	3.066839	4.374703	1.307864	0.009393
20	4.782106	5.877438	1.095332	0.000115

Double DQN

SNR	Agent	Oracle	Gap	BER
-10	0.039155	0.082367	0.043212	0.412909
-5	0.169297	0.254247	0.084950	0.321580
0	0.422477	0.708215	0.285738	0.223599
5	0.886671	1.520290	0.633619	0.117326
10	2.012694	2.805952	0.793258	0.037526
15	3.012210	4.374703	1.362493	0.022504
20	4.627910	5.877438	1.249529	0.000332

Dueling DQN

SNR	Agent	Oracle	Gap	BER
-10	0.057057	0.082367	0.025309	0.391246
-5	0.155709	0.254247	0.098538	0.325028
0	0.439450	0.708215	0.268765	0.210721
5	1.009602	1.520290	0.510688	0.097121
10	2.324150	2.805952	0.481802	0.008809
15	3.241224	4.374703	1.133479	0.007987
20	4.984020	5.877438	0.893418	0.003010

5. Interpretation and limitations

- All three algorithms completed successfully in the executed seed-42 run.
- The Dueling DQN row has the largest final-rate mean in this single verification run; this is descriptive only and is not a statistical superiority claim.
- The SNR sweep compares agent spectral efficiency with an exhaustive finite-codebook oracle on fresh channels.
- BER is an analytical BPSK/AWGN estimate rather than a full link-level waveform simulation.
- The run uses 100 episodes and one seed, so it does not provide multi-seed confidence intervals or reproduce the manuscript's 1000-episode reference campaign.